PROJECT OVERVIEW

Business Analytics

A toy LLM-assisted tool that answers questions about stocks and other financial entities

Rose Nestor · Springboard “AI for Programmers” Bootcamp · Fall 2026

What it does

You ask

“Should I invest in [company name]?”

Typed into the command line (main.py) or a simple desktop window (gui.py).

It answers

OpenAI’s GPT-4o mini, prompted as a financial analyst, forecasts the entity’s performance using trends from social media and prediction markets.

Sources named in the prompt: X, Kalshi, Bluesky, Polymarket, Instagram, Google.

How it works

1

Ask

User types a free-form question

→

2

Parse

LLM fills a Pydantic **Query**: entities + questions

→

3

Confirm

“Did you ask…?” (y/n), one retry if not

→

4

Forecast

Second LLM call answers the confirmed query

Every LLM call is **traced in LangSmith** and logged to **main.log**, then the loop repeats until the user presses q.

Bootcamp concepts it demonstrates

Structured output

responses.parse returns a validated Pydantic model

Chaining + human-in-the-loop

Two linked calls with a human check in between

Traceability & observability

LangSmith @traceable decorators, wrapped OpenAI client, logging

Long-lived context with skills

AGENTS.md plus skills for code review, commits, docs and tests

Also scaffolded: a RAG pipeline (chunk → embed → Pinecone search) as a planned extension.

Tech stack

Tool	Role in the project
OpenAI API	GPT-4o mini for parsing and forecasting
Pydantic	Schema for the structured Query
LangSmith	Tracing and observability
Pinecone	Vector store for RAG (planned)
PyQt6	Desktop GUI prototype
Python 3	Platform-independent · Apache-2.0 licence

Where it stands (v0.1)

Working

- Command-line question loop
- Structured parsing with Pydantic
- Human confirmation step
- LangSmith tracing and logging

In progress

- RAG / embeddings code not yet wired in
- GUI is a placeholder, untested, no HITL
- Test suite still empty
- Package refactor for test.pypi.org

What’s next

Smarter answers

RAG over Nasdaq data with Pinecone; Anthropic as a second provider

Better interface

Finish the GUI with HITL, ship an executable, maybe a curses CLI; export to Excel or JSON

Engineering

Tests and profiling, PyPI packaging, PyO3/Cython rewrites with benchmarks

Questions?

github.com/RuminativeRose/business_analytics2